Chapter 5

General Minimum Variance Unbiased Estimation

5.1 Introduction

We have seen that the evaluation of the CRLB sometimes results in an efficient and hence MVU estimator. In particular, the linear model provides a useful example of this approach. If, however, an efficient estimator does not exist, it is still of interest to be able to find the MVU estimator (assuming of course that it exists). To do so requires the concept of sufficient statistics and the important Rao-Blackwell-Lehmann-Scheffe theorem. Armed with this theory it is possible in many cases to determine the MVU estimator by a simple inspection of the PDF. How this is done is explored in this chapter.

5.2 Summary

In Section 5.3 we define a sufficient statistic as one which summarizes the data in the sense that if we are given the sufficient statistic, the PDF of the data no longer depends on the unknown parameter. The Neyman-Fisher factorization theorem (see Theorem 5.1) enables us to easily find sufficient statistics by examining the PDF. Once the sufficient statistic has been found, the MVU estimator may be determined by finding a function of the sufficient statistic that is an unbiased estimator. The essence of the approach is summarized in Theorem 5.2 and is known as the Rao-Blackwell-Lehmann-Scheffe theorem. In using this theorem we must also ensure that the sufficient statistic is complete or that there exists only one function of it that is an unbiased estimator. The extension to the vector parameter case is given in Theorems 5.3 and 5.4. The approach is basically the same as for a scalar parameter. We must first find as many sufficient statistics as unknown parameters and then determine an unbiased estimator based on the sufficient statistics.

5.3 Sufficient Statistics

In a previous chapter we found that for the problem of estimating a DC level A in WGN (see Example 3.3), the sample mean

$$\hat{A} = \frac{1}{N}\sum_{n=0}^{N-1} x[n]$$

was the MVU estimator, having minimum variance σ^2/N. If, on the other hand, we had chosen

$$\check{A} = x[0]$$

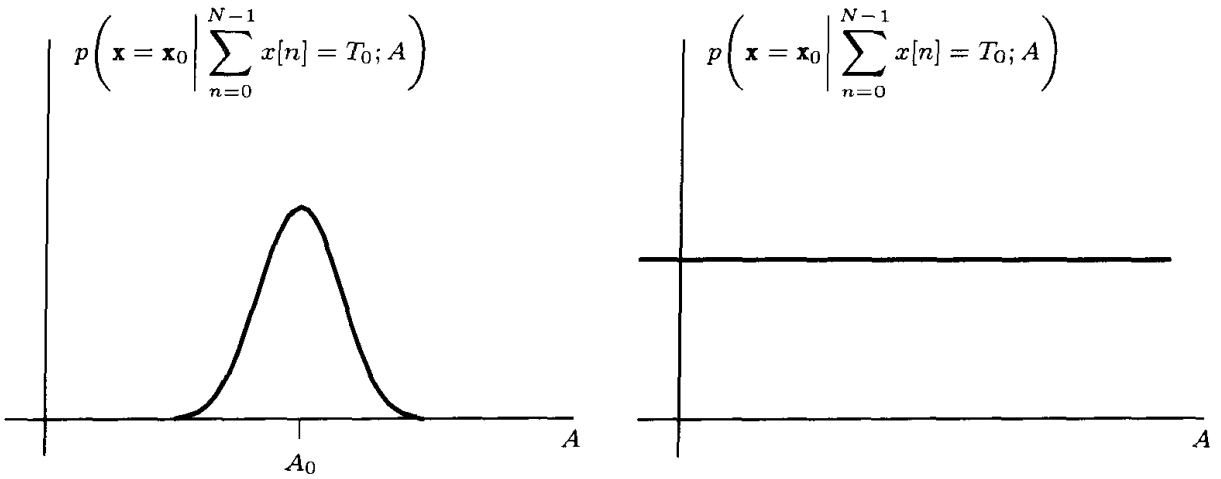
as our estimator, it is immediately clear that even though $\check{A}$ is unbiased, its variance is much larger (being σ^2) than the minimum. Intuitively, the poor performance is a direct result of discarding the data points $\{x[1], x[2], \ldots, x[N-1]\}$ which carry information about A. A reasonable question to ask is Which data samples are pertinent to the estimation problem? or Is there a set of data that is sufficient?. The following data sets may be claimed to be sufficient in that they may be used to compute $\hat{A}$.

$$\begin{aligned} S_1 &= \{x[0], x[1], \ldots, x[N-1]\} \\ S_2 &= \{x[0]+x[1], x[2], x[3], \ldots, x[N-1]\} \\ S_3 &= \left\{\sum_{n=0}^{N-1} x[n]\right\}. \end{aligned}$$

S_1 represents the original data set, which as expected, is always sufficient for the problem. S_2 and S_3 are also sufficient. It is obvious that for this problem there are many sufficient data sets. The data set that contains the least number of elements is called the *minimal* one. If we now think of the elements of these sets as statistics, we say that the N statistics of S_1 are *sufficient*, as well as the $(N-1)$ statistics of S_2 and the single statistic of S_3. This latter statistic, $\sum_{n=0}^{N-1} x[n]$, in addition to being a *sufficient statistic*, is the *minimal sufficient statistic*. For estimation of A, once we know $\sum_{n=0}^{N-1} x[n]$, we no longer need the individual data values since all information has been summarized in the sufficient statistic. To quantify what we mean by this, consider the PDF of the data

$$p(\mathbf{x}; A) = \frac{1}{(2\pi\sigma^2)^{\frac{N}{2}}} \exp\left[-\frac{1}{2\sigma^2}\sum_{n=0}^{N-1}(x[n]-A)^2\right] \tag{5.1}$$

and assume that $T(\mathbf{x}) = \sum_{n=0}^{N-1} x[n] = T_0$ has been observed. Knowledge of the value of this statistic will change the PDF to the conditional one $p(\mathbf{x}|\sum_{n=0}^{N-1} x[n] = T_0; A)$, which now gives the PDF of the observations after the sufficient statistic has been observed. Since the statistic is sufficient for the estimation of A, this conditional PDF should not



(a) Observations provide information after $T(\mathbf{x})$ observed—$T(\mathbf{x})$ is not sufficient

(b) No information from observations after $T(\mathbf{x})$ observed—$T(\mathbf{x})$ is sufficient

Figure 5.1 Sufficient statistic definition

depend on A. If it did, then we could infer some additional information about A from the data in addition to that already provided by the sufficient statistic. As an example, in Figure 5.1a, if $\mathbf{x} = \mathbf{x}_0$ for an arbitrary $\mathbf{x}_0$, then values of A near A_0 would be more likely. This violates our notion that $\sum_{n=0}^{N-1} x[n]$ is a sufficient statistic. On the other hand, in Figure 5.1b, any value of A is as likely as any other, so that after observing $T(\mathbf{x})$ the data may be discarded. Hence, to verify that a statistic is sufficient we need to determine the conditional PDF and confirm that there is no dependence on A.

Example 5.1 - Verification of a Sufficient Statistic

Consider the PDF of (5.1). To prove that $\sum_{n=0}^{N-1} x[n]$ is a sufficient statistic we need to determine $p(\mathbf{x}|T(\mathbf{x}) = T_0; A)$, where $T(\mathbf{x}) = \sum_{n=0}^{N-1} x[n]$. By the definition of the conditional PDF we have

$$p(\mathbf{x}|T(\mathbf{x}) = T_0; A) = \frac{p(\mathbf{x}, T(\mathbf{x}) = T_0; A)}{p(T(\mathbf{x}) = T_0; A)}.$$

But note that $T(\mathbf{x})$ is functionally dependent on $\mathbf{x}$, so that the *joint* PDF $p(\mathbf{x}, T(\mathbf{x}) = T_0; A)$ takes on nonzero values only when $\mathbf{x}$ satisfies $T(\mathbf{x}) = T_0$. The joint PDF is therefore $p(\mathbf{x}; A)\delta(T(\mathbf{x}) - T_0)$, where δ is the Dirac delta function (see also Appendix 5A for a further discussion). Thus, we have that

$$p(\mathbf{x}|T(\mathbf{x}) = T_0; A) = \frac{p(\mathbf{x}; A)\delta(T(\mathbf{x}) - T_0)}{p(T(\mathbf{x}) = T_0; A)}. \tag{5.2}$$

Clearly, $T(\mathbf{x}) \sim \mathcal{N}(NA, N\sigma^2)$, so that

$$p(\mathbf{x}; A)\delta(T(\mathbf{x}) - T_0)$$

$$\begin{aligned}
&= \frac{1}{(2\pi\sigma^2)^{\frac{N}{2}}} \exp\left[-\frac{1}{2\sigma^2}\sum_{n=0}^{N-1}(x[n]-A)^2\right]\delta(T(\mathbf{x})-T_0) \\
&= \frac{1}{(2\pi\sigma^2)^{\frac{N}{2}}} \exp\left[-\frac{1}{2\sigma^2}\left(\sum_{n=0}^{N-1}x^2[n] - 2AT(\mathbf{x}) + NA^2\right)\right]\delta(T(\mathbf{x})-T_0) \\
&= \frac{1}{(2\pi\sigma^2)^{\frac{N}{2}}} \exp\left[-\frac{1}{2\sigma^2}\left(\sum_{n=0}^{N-1}x^2[n] - 2AT_0 + NA^2\right)\right]\delta(T(\mathbf{x})-T_0).
\end{aligned}$$

From (5.2) we have

$$p(\mathbf{x}|T(\mathbf{x}) = T_0; A)$$

$$\begin{aligned}
&= \frac{\frac{1}{(2\pi\sigma^2)^{\frac{N}{2}}} \exp\left[-\frac{1}{2\sigma^2}\sum_{n=0}^{N-1}x^2[n]\right]\exp\left[-\frac{1}{2\sigma^2}(-2AT_0+NA^2)\right]}{\frac{1}{\sqrt{2\pi N\sigma^2}}\exp\left[-\frac{1}{2N\sigma^2}(T_0-NA)^2\right]}\delta(T(\mathbf{x})-T_0) \\
&= \frac{\sqrt{N}}{(2\pi\sigma^2)^{\frac{N-1}{2}}} \exp\left[-\frac{1}{2\sigma^2}\sum_{n=0}^{N-1}x^2[n]\right]\exp\left[\frac{T_0^2}{2N\sigma^2}\right]\delta(T(\mathbf{x})-T_0)
\end{aligned}$$

which as claimed does not depend on A. Therefore, we can conclude that $\sum_{n=0}^{N-1} x[n]$ is a sufficient statistic for the estimation of A. ◇

This example indicates the procedure for verifying that a statistic is sufficient. For many problems the task of evaluating the conditional PDF is formidable, so that an easier approach is needed. Additionally, in Example 5.1 the choice of $\sum_{n=0}^{N-1} x[n]$ for examination as a sufficient statistic was fortuitous. In general an even more difficult problem would be to *identify* potential sufficient statistics. The approach of guessing at a sufficient statistic and then verifying it is, of course, quite unsatisfactory in practice. To alleviate the guesswork we can employ the Neyman-Fisher factorization theorem, which is a simple "turn-the-crank" procedure for finding sufficient statistics.

5.4 Finding Sufficient Statistics

The Neyman-Fisher factorization theorem is now stated, after which we will use it to find sufficient statistics in several examples.

Theorem 5.1 (Neyman-Fisher Factorization) *If we can factor the PDF* $p(\mathbf{x}; \theta)$ *as*

$$p(\mathbf{x}; \theta) = g(T(\mathbf{x}), \theta)h(\mathbf{x}) \tag{5.3}$$

where g is a function depending on $\mathbf{x}$ only through $T(\mathbf{x})$ and h is a function depending only on $\mathbf{x}$, then $T(\mathbf{x})$ is a sufficient statistic for θ. Conversely, if $T(\mathbf{x})$ is a sufficient statistic for θ, then the PDF can be factored as in (5.3).

A proof of this theorem is contained in Appendix 5A. It should be mentioned that at times it is not obvious if the PDF can be factored in the required form. If this is the case, then a sufficient statistic may not exist. Some examples are now given to illustrate the use of this powerful theorem.

Example 5.2 - DC Level in WGN

We now reexamine the problem discussed in the previous section. There the PDF was given by (5.1), where we note that σ^2 is assumed known. To demonstrate that a factorization exists we observe that the exponent of the PDF may be rewritten as

$$\sum_{n=0}^{N-1}(x[n]-A)^2 = \sum_{n=0}^{N-1}x^2[n] - 2A\sum_{n=0}^{N-1}x[n] + NA^2$$

so that the PDF is factorable as

$$p(\mathbf{x};A) = \underbrace{\frac{1}{(2\pi\sigma^2)^{\frac{N}{2}}}\exp\left[-\frac{1}{2\sigma^2}\left(NA^2 - 2A\sum_{n=0}^{N-1}x[n]\right)\right]}_{g(T(\mathbf{x}),A)}\underbrace{\exp\left[-\frac{1}{2\sigma^2}\sum_{n=0}^{N-1}x^2[n]\right]}_{h(\mathbf{x})}.$$

Clearly then, $T(\mathbf{x}) = \sum_{n=0}^{N-1}x[n]$ is a sufficient statistic for A. Note that $T'(\mathbf{x}) = 2\sum_{n=0}^{N-1}x[n]$ is also a sufficient statistic for A, and in fact *any one-to-one function of* $\sum_{n=0}^{N-1}x[n]$ *is a sufficient statistic* (see Problem 5.12). Hence, sufficient statistics are unique only to within one-to-one transformations. ◇

Example 5.3 - Power of WGN

Now consider the PDF of (5.1) with $A = 0$ and σ^2 as the unknown parameter. Then,

$$p(\mathbf{x};\sigma^2) = \underbrace{\frac{1}{(2\pi\sigma^2)^{\frac{N}{2}}}\exp\left[-\frac{1}{2\sigma^2}\sum_{n=0}^{N-1}x^2[n]\right]}_{g(T(\mathbf{x}),\sigma^2)}\cdot\underbrace{1}_{h(\mathbf{x})}.$$

Again it is immediately obvious from the factorization theorem that $T(\mathbf{x}) = \sum_{n=0}^{N-1}x^2[n]$ is a sufficient statistic for σ^2. See also Problem 5.1. ◇

Example 5.4 - Phase of Sinusoid

Recall the problem in Example 3.4 in which we wish to estimate the phase of a sinusoid embedded in WGN or

$$x[n] = A\cos(2\pi f_0 n + \phi) + w[n] \qquad n = 0, 1, \ldots, N-1.$$

Here, the amplitude A and frequency f_0 of the sinusoid are known, as is the noise variance σ^2. The PDF is

$$p(\mathbf{x};\phi) = \frac{1}{(2\pi\sigma^2)^{\frac{N}{2}}} \exp\left\{-\frac{1}{2\sigma^2}\sum_{n=0}^{N-1}\left[x[n] - A\cos(2\pi f_0 n + \phi)\right]^2\right\}.$$

The exponent may be expanded as

$$\sum_{n=0}^{N-1} x^2[n] - 2A\sum_{n=0}^{N-1} x[n]\cos(2\pi f_0 n + \phi) + \sum_{n=0}^{N-1} A^2\cos^2(2\pi f_0 n + \phi)$$

$$\begin{aligned} = & \sum_{n=0}^{N-1} x^2[n] - 2A\left(\sum_{n=0}^{N-1} x[n]\cos 2\pi f_0 n\right)\cos\phi \\ & + 2A\left(\sum_{n=0}^{N-1} x[n]\sin 2\pi f_0 n\right)\sin\phi + \sum_{n=0}^{N-1} A^2\cos^2(2\pi f_0 n + \phi). \end{aligned}$$

In this example it does not appear that the PDF is factorable as required by the Neyman-Fisher theorem. Hence, no *single* sufficient statistic exists. However, it can be factored as

$$p(\mathbf{x};\phi) =$$

$$\underbrace{\frac{1}{(2\pi\sigma^2)^{\frac{N}{2}}} \exp\left\{-\frac{1}{2\sigma^2}\left[\sum_{n=0}^{N-1} A^2\cos^2(2\pi f_0 n + \phi) - 2AT_1(\mathbf{x})\cos\phi + 2AT_2(\mathbf{x})\sin\phi\right]\right\}}_{g(T_1(\mathbf{x}), T_2(\mathbf{x}), \phi)}$$

$$\underbrace{\cdot \exp\left[-\frac{1}{2\sigma^2}\sum_{n=0}^{N-1} x^2[n]\right]}_{h(\mathbf{x})}$$

where

$$\begin{aligned} T_1(\mathbf{x}) &= \sum_{n=0}^{N-1} x[n]\cos 2\pi f_0 n \\ T_2(\mathbf{x}) &= \sum_{n=0}^{N-1} x[n]\sin 2\pi f_0 n. \end{aligned}$$

By a slight generalization of the Neyman-Fisher theorem we can conclude that $T_1(\mathbf{x})$ *and* $T_2(\mathbf{x})$ are *jointly* sufficient statistics for the estimation of ϕ. However, no *single* sufficient statistic exists. The reason why we wish to restrict our attention to single sufficient statistics will become clear in the next section. $\diamond$

The concept of jointly sufficient statistics is a simple extension of our previous definition. The r statistics $T_1(\mathbf{x}), T_2(\mathbf{x}), \ldots, T_r(\mathbf{x})$ are jointly sufficient statistics if the conditional PDF $p(\mathbf{x}|T_1(\mathbf{x}), T_2(\mathbf{x}), \ldots, T_r(\mathbf{x}); \theta)$ does not depend on θ. The generalization of the Neyman-Fisher theorem asserts that if $p(\mathbf{x}; \theta)$ can be factored as [Kendall and Stuart 1979]

$$p(\mathbf{x}; \theta) = g(T_1(\mathbf{x}), T_2(\mathbf{x}), \ldots, T_r(\mathbf{x}), \theta)h(\mathbf{x}) \tag{5.4}$$

then $\{T_1(\mathbf{x}), T_2(\mathbf{x}), \ldots, T_r(\mathbf{x})\}$ are sufficient statistics for θ. It is clear then that the original data are always sufficient statistics since we can let $r = N$ and

$$T_{n+1}(\mathbf{x}) = x[n] \qquad n = 0, 1, \ldots, N-1$$

so that

$$\begin{aligned} g &= p(\mathbf{x}; \theta) \\ h &= 1 \end{aligned}$$

and (5.4) holds identically. Of course, they are seldom the *minimal* set of sufficient statistics.

5.5 Using Sufficiency to Find the MVU Estimator

Assuming that we have been able to find a sufficient statistic $T(\mathbf{x})$ for θ, we can make use of the Rao-Blackwell-Lehmann-Scheffe (RBLS) theorem to find the MVU estimator. We will first illustrate the approach with an example and then state the theorem formally.

Example 5.5 - DC Level in WGN

We will continue Example 5.2. Although we already know that $\hat{A} = \bar{x}$ is the MVU estimator (since it is efficient), we will use the RBLS theorem, which can be used even when an efficient estimator does not exist and hence the CRLB method is no longer viable. The procedure for finding the MVU estimator $\hat{A}$ may be implemented in two different ways. They are both based on the sufficient statistic $T(\mathbf{x}) = \sum_{n=0}^{N-1} x[n]$.

1. Find *any* unbiased estimator of A, say $\check{A} = x[0]$, and determine $\hat{A} = E(\check{A}|T)$. The expectation is taken with respect to $p(\check{A}|T)$.

2. Find some function g so that $\hat{A} = g(T)$ is an unbiased estimator of A.

For the first approach we can let the unbiased estimator be $\check{A} = x[0]$ and determine $\hat{A} = E(x[0]|\sum_{n=0}^{N-1} x[n])$. To do so we will need some properties of the conditional

Gaussian PDF. For $[x\,y]^T$ a Gaussian random vector with mean vector $\boldsymbol{\mu} = [E(x)\,E(y)]^T$ and covariance matrix

$$\mathbf{C} = \begin{bmatrix} \text{var}(x) & \text{cov}(x,y) \\ \text{cov}(y,x) & \text{var}(y) \end{bmatrix},$$

it may be shown that (see Appendix 10A)

$$\begin{aligned} E(x|y) &= \int_{-\infty}^{\infty} xp(x|y)\,dx \\ &= \int_{-\infty}^{\infty} x\frac{p(x,y)}{p(y)}\,dx \\ &= E(x) + \frac{\text{cov}(x,y)}{\text{var}(y)}(y - E(y)). \end{aligned} \tag{5.5}$$

Applying this result, we let $x = x[0]$ and $y = \sum_{n=0}^{N-1} x[n]$ and note that

$$\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} x[0] \\ \sum_{n=0}^{N-1} x[n] \end{bmatrix} = \underbrace{\begin{bmatrix} 1 & 0 & 0 & \dots & 0 \\ 1 & 1 & 1 & \dots & 1 \end{bmatrix}}_{\mathbf{L}} \begin{bmatrix} x[0] \\ x[1] \\ \vdots \\ x[N-1] \end{bmatrix}.$$

Hence, the PDF of $[x\,y]^T$ is $\mathcal{N}(\boldsymbol{\mu}, \mathbf{C})$ since this represents a linear transformation of a Gaussian vector, where

$$\begin{aligned} \boldsymbol{\mu} &= \mathbf{L}E(\mathbf{x}) = \mathbf{L}A\mathbf{1} = \begin{bmatrix} A \\ NA \end{bmatrix} \\ \mathbf{C} &= \sigma^2\mathbf{L}\mathbf{L}^T = \sigma^2\begin{bmatrix} 1 & 1 \\ 1 & N \end{bmatrix}. \end{aligned}$$

Hence, we have finally from (5.5) that

$$\begin{aligned} \hat{A} &= E(x|y) = A + \frac{\sigma^2}{N\sigma^2}\left(\sum_{n=0}^{N-1} x[n] - NA\right) \\ &= \frac{1}{N}\sum_{n=0}^{N-1} x[n] \end{aligned}$$

which is the MVU estimator. This approach, requiring evaluation of a conditional expectation, is usually mathematically intractable.

Turning our attention to the second approach, we need to find some function g so that

$$\hat{A} = g\left(\sum_{n=0}^{N-1} x[n]\right)$$

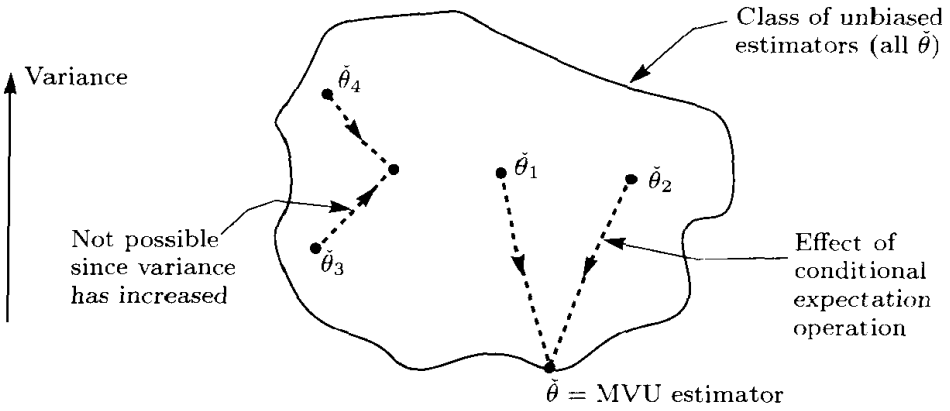


Figure 5.2 RBLS argument for MVU estimator

is an unbiased estimator of A. By inspection this is $g(x) = x/N$, which yields

$$\hat{A} = \frac{1}{N}\sum_{n=0}^{N-1} x[n]$$

as the MVU estimator. This alternative method is much easier to apply, and therefore in practice, it is the one we generally employ. ◇

We now formally state the RBLS theorem.

Theorem 5.2 (Rao-Blackwell-Lehmann-Scheffe) *If $\check{\theta}$ is an unbiased estimator of θ and $T(\mathbf{x})$ is a sufficient statistic for θ, then $\hat{\theta} = E(\check{\theta}|T(\mathbf{x}))$ is*

1. *a valid estimator for θ (not dependent on θ)*
2. *unbiased*
3. *of lesser or equal variance than that of $\check{\theta}$, for all θ.*

Additionally, if the sufficient statistic is complete, then $\hat{\theta}$ is the MVU estimator.

A proof is given in Appendix 5B. In the previous example we saw that $E(x[0]|\sum_{n=0}^{N-1} x[n]) = \bar{x}$ did not depend on A, making it a valid estimator, was unbiased, and had less variance than $x[0]$. That there is no other estimator with less variance, as Theorem 5.2 asserts, follows from the property that the sufficient statistic $\sum_{n=0}^{N-1} x[n]$ is a *complete* sufficient statistic. In essence, a statistic is complete if *there is only one function of the statistic that is unbiased.* The argument that $\hat{\theta} = E(\check{\theta}|T(\mathbf{x}))$ is the MVU estimator is now given. Consider all possible unbiased estimators of θ, as depicted in Figure 5.2. By determining $E(\check{\theta}|T(\mathbf{x}))$ we can lower the variance of the estimator (property 3 of Theorem 5.2) and still remain within the class (property 2 of Theorem 5.2). But $E(\check{\theta}|T(\mathbf{x}))$ is solely a function of the sufficient statistic $T(\mathbf{x})$ since

$$\begin{aligned}\hat{\theta} &= E(\check{\theta}|T(\mathbf{x})) = \int \check{\theta} p(\check{\theta}|T(\mathbf{x}))\, d\check{\theta} \\ &= g(T(\mathbf{x})). \end{aligned} \tag{5.6}$$

If $T(\mathbf{x})$ is complete, there is but *one* function of T that is an unbiased estimator. Hence, $\hat{\theta}$ is *unique*, independent of the $\check{\theta}$ we choose from the class shown in Figure 5.2. Every $\check{\theta}$ maps into the *same estimator* $\hat{\theta}$. Because the variance of $\hat{\theta}$ must be less than $\check{\theta}$ for any $\check{\theta}$ within the class (property 3 of Theorem 5.2), we conclude that $\hat{\theta}$ must be the MVU estimator. In summary, the MVU estimator can be found by taking *any* unbiased estimator and carrying out the operations in (5.6). Alternatively, since there is only one function of the sufficient statistic that leads to an unbiased estimator, we need only find the unique g to make the sufficient statistic unbiased. For this latter approach we found in Example 5.5 that $g(\sum_{n=0}^{N-1} x[n]) = \sum_{n=0}^{N-1} x[n]/N$.

The property of completeness depends on the PDF of $\mathbf{x}$, which in turn determines the PDF of the sufficient statistic. For many practical cases of interest it holds. In particular, for the exponential family of PDFs (see Problems 5.14 and 5.15) this condition is satisfied. To validate that a sufficient statistic is complete is in general quite difficult, and we refer the reader to the discussions in [Kendall and Stuart 1979]. A flavor for the concept of completeness is provided by the next two examples.

Example 5.6 - Completeness of a Sufficient Statistic

For the estimation of A, the sufficient statistic $\sum_{n=0}^{N-1} x[n]$ is complete or there is but one function g for which $E[g(\sum_{n=0}^{N-1} x[n])] = A$. Suppose, however, that there exists a second function h for which $E[h(\sum_{n=0}^{N-1} x[n])] = A$. Then, it would follow that with $T = \sum_{n=0}^{N-1} x[n]$,

$$E\left[g(T) - h(T)\right] = A - A = 0 \qquad \text{for all } A$$

or since $T \sim \mathcal{N}(NA, N\sigma^2)$

$$\int_{-\infty}^{\infty} v(T) \frac{1}{\sqrt{2\pi N\sigma^2}} \exp\left[-\frac{1}{2N\sigma^2}(T - NA)^2\right] dT = 0 \qquad \text{for all } A$$

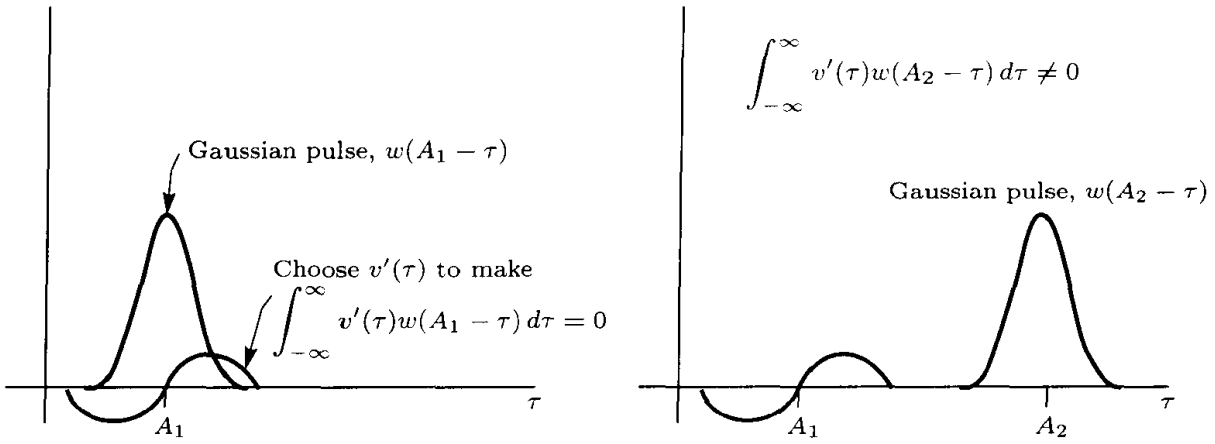
where $v(T) = g(T) - h(T)$. Letting $\tau = T/N$ and $v'(\tau) = v(N\tau)$, we have

$$\int_{-\infty}^{\infty} v'(\tau) \frac{N}{\sqrt{2\pi N\sigma^2}} \exp\left[-\frac{N}{2\sigma^2}(A - \tau)^2\right] d\tau = 0 \qquad \text{for all } A \tag{5.7}$$

which may be recognized as the convolution of a function $v'(\tau)$ with a Gaussian pulse $w(\tau)$ (see Figure 5.3). For the result to be zero for all A, $v'(\tau)$ must be identically zero. To see this recall that a signal is zero if and only if its Fourier transform is identically zero, resulting in the condition

$$V'(f)W(f) = 0 \qquad \text{for all } f$$

where $V'(f) = \mathcal{F}\{v'(\tau)\}$ and $W(f)$ is the Fourier transform of the Gaussian pulse in (5.7). Since $W(f)$ is also Gaussian and therefore positive for all f, we have that the condition is satisfied if and only if $V'(f) = 0$ for all f. Hence, we must have that $v'(\tau) = 0$ for all τ. This implies that $g = h$ or that the function g is unique. $\diamond$



(a) Integral equals zero **(b)** Integral does not equal zero

Figure 5.3 Completeness condition for sufficient statistic (satisfied)

Example 5.7 - Incomplete Sufficient Statistic

Consider the estimation of A for the datum

$$x[0] = A + w[0]$$

where $w[0] \sim \mathcal{U}[-\frac{1}{2}, \frac{1}{2}]$. A sufficient statistic is $x[0]$, being the only available data, and furthermore, $x[0]$ is an unbiased estimator of A. We may conclude that $g(x[0]) = x[0]$ is a viable candidate for the MVU estimator. That it is actually the MVU estimator still requires us to verify that it is a complete sufficient statistic. As in the previous example, we suppose that there exists another function h with the unbiased property $h(x[0]) = A$ and attempt to prove that $h = g$. Again letting $v(T) = g(T) - h(T)$, we examine the possible solutions for v of the equation

$$\int_{-\infty}^{\infty} v(T)p(\mathbf{x}; A)\,d\mathbf{x} = 0 \qquad \text{for all } A.$$

For this problem, however, $\mathbf{x} = x[0] = T$, so that

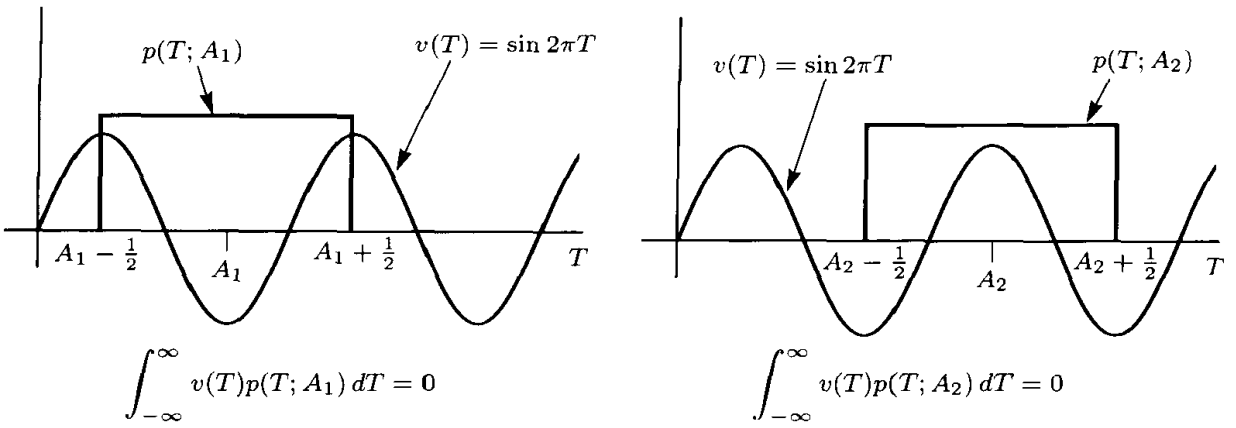
$$\int_{-\infty}^{\infty} v(T)p(T; A)\,dT = 0 \qquad \text{for all A.}$$

But

$$p(T; A) = \begin{cases} 1 & A - \frac{1}{2} \leq T \leq A + \frac{1}{2} \\ 0 & \text{otherwise} \end{cases}$$

so that the condition reduces to

$$\int_{A-\frac{1}{2}}^{A+\frac{1}{2}} v(T)\,dT = 0 \qquad \text{for all } A.$$



(a) Integral equals zero **(b)** Integral equals zero

Figure 5.4 Completeness condition for sufficient statistic (not satisfied)

The nonzero function $v(T) = \sin 2\pi T$ will satisfy this condition as illustrated in Figure 5.4. Hence a solution is

$$v(T) = g(T) - h(T) = \sin 2\pi T$$

or

$$h(T) = T - \sin 2\pi T.$$

As a result, the estimator

$$\hat{A} = x[0] - \sin 2\pi x[0]$$

is also based on the sufficient statistic and is unbiased for A. Having found at least one other unbiased estimator that is also a function of the sufficient statistic, we may conclude that the sufficient statistic is not complete. The RBLS theorem no longer holds, and it is not possible to assert that $\hat{A} = x[0]$ is the MVU estimator. $\diamond$

To summarize the completeness condition, we say that a sufficient statistic is complete if the condition

$$\int_{-\infty}^{\infty} v(T)p(T; \theta)\,dT = 0 \qquad \text{for all } \theta \tag{5.8}$$

is *satisfied only by the zero function* or by $v(T) = 0$ for all T.

At this point it is worthwhile to review our results and then apply them to an estimation problem for which we do not know the MVU estimator. The procedure is as follows (see also Figure 5.5):

1. Find a single sufficient statistic for θ, that is, $T(\mathbf{x})$, by using the Neyman-Fisher factorization theorem.

2. Determine if the sufficient statistic is complete and, if so, proceed; if not, this approach cannot be used.

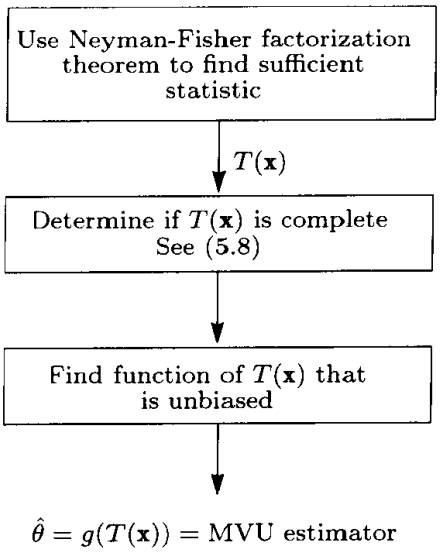


Figure 5.5 Procedure for finding MVU estimator (scalar parameter)

3. Find a function g of the sufficient statistic that yields an unbiased estimator $\hat{\theta} = g(T(\mathbf{x}))$. The MVU estimator is then $\hat{\theta}$.

As an alternative implementation of step 3 we may

3.′ Evaluate $\hat{\theta} = E(\check{\theta}|T(\mathbf{x}))$, where $\check{\theta}$ is any unbiased estimator.

However, in practice the conditional expectation evaluation is usually too tedious. The next example illustrates the overall procedure.

Example 5.8 - Mean of Uniform Noise

We observe the data

$$x[n] = w[n] \qquad n = 0, 1, \ldots, N-1$$

where $w[n]$ is IID noise with PDF $\mathcal{U}[0, \beta]$ for $\beta > 0$. We wish to find the MVU estimator for the mean $\theta = \beta/2$. Our initial approach of using the CRLB for finding an efficient and hence MVU estimator cannot even be tried for this problem. This is because the PDF does not satisfy the required regularity conditions (see Problem 3.1). A natural estimator of θ is the sample mean or

$$\hat{\theta} = \frac{1}{N} \sum_{n=0}^{N-1} x[n].$$

The sample mean is easily shown to be unbiased and to have variance

$$\begin{aligned} \text{var}(\hat{\theta}) &= \frac{1}{N} \text{var}(x[n]) \\ &= \frac{\beta^2}{12N}. \end{aligned} \tag{5.9}$$

To determine if the sample mean is the MVU estimator for this problem we will follow the procedure outlined in Figure 5.5. Define the unit step function as

$$u(x) = \begin{cases} 1 & \text{for } x > 0 \\ 0 & \text{for } x < 0. \end{cases}$$

Then,

$$p(x[n];\theta) = \frac{1}{\beta}\left[u(x[n]) - u(x[n]-\beta)\right]$$

where $\beta = 2\theta$, and therefore the PDF of the data is

$$p(\mathbf{x};\theta) = \frac{1}{\beta^N}\prod_{n=0}^{N-1}\left[u(x[n]) - u(x[n]-\beta)\right].$$

This PDF will be nonzero only if $0 < x[n] < \beta$ for all $x[n]$, so that

$$p(\mathbf{x};\theta) = \begin{cases} \frac{1}{\beta^N} & 0 < x[n] < \beta \quad n = 0, 1, \ldots, N-1 \\ 0 & \text{otherwise.} \end{cases}$$

Alternatively, we can write

$$p(\mathbf{x};\theta) = \begin{cases} \frac{1}{\beta^N} & \max x[n] < \beta,\ \min x[n] > 0 \\ 0 & \text{otherwise} \end{cases}$$

so that

$$p(\mathbf{x};\theta) = \underbrace{\frac{1}{\beta^N}u(\beta - \max x[n])}_{g(T(\mathbf{x}),\theta)}\underbrace{u(\min x[n])}_{h(\mathbf{x})}.$$

By the Neyman-Fisher factorization theorem $T(\mathbf{x}) = \max x[n]$ is a sufficient statistic for θ. Furthermore, it can be shown that the sufficient statistic is complete. We omit the proof.

Next we need to determine a function of the sufficient statistic to make it unbiased. To do so requires us to determine the expected value of $T = \max x[n]$. The statistic T is known as an *order statistic*. Its PDF is now derived. Evaluating first the cumulative distribution function

$$\begin{aligned} \Pr\{T \le \xi\} &= \Pr\{x[0] \le \xi, x[1] \le \xi, \ldots, x[N-1] \le \xi\} \\ &= \prod_{n=0}^{N-1}\Pr\{x[n] \le \xi\} \\ &= \Pr\{x[n] \le \xi\}^N \end{aligned}$$

since the random variables are IID. The PDF follows as

$$\begin{aligned} p_T(\xi) &= \frac{d\Pr\{T \leq \xi\}}{d\xi} \\ &= N\Pr\{x[n] \leq \xi\}^{N-1}\frac{d\Pr\{x[n] \leq \xi\}}{d\xi}. \end{aligned}$$

But $d\Pr\{x[n] \leq \xi\}/d\xi$ is the PDF of $x[n]$ or

$$p_{x[n]}(\xi;\theta) = \begin{cases} \frac{1}{\beta} & 0 < \xi < \beta \\ 0 & \text{otherwise.} \end{cases}$$

Integrating, we obtain

$$\Pr\{x[n] \leq \xi\} = \begin{cases} 0 & \xi < 0 \\ \frac{\xi}{\beta} & 0 < \xi < \beta \\ 1 & \xi > \beta \end{cases}$$

which finally yields

$$p_T(\xi) = \begin{cases} 0 & \xi < 0 \\ N\left(\frac{\xi}{\beta}\right)^{N-1}\frac{1}{\beta} & 0 < \xi < \beta \\ 0 & \xi > \beta. \end{cases}$$

We now have

$$\begin{aligned} E(T) &= \int_{-\infty}^{\infty} \xi p_T(\xi)\, d\xi \\ &= \int_0^{\beta} \xi N\left(\frac{\xi}{\beta}\right)^{N-1}\frac{1}{\beta}\, d\xi \\ &= \frac{N}{N+1}\beta \\ &= \frac{2N}{N+1}\theta. \end{aligned}$$

To make this unbiased we let $\hat{\theta} = [(N+1)/2N]T$, so that finally the MVU estimator is

$$\hat{\theta} = \frac{N+1}{2N}\max x[n].$$

Somewhat contrary to intuition, the sample mean is not the MVU estimator of the mean for uniformly distributed noise! It is of interest to compare the minimum variance to that of the sample mean estimator variance. Noting that

$$\text{var}(\hat{\theta}) = \left(\frac{N+1}{2N}\right)^2 \text{var}(T)$$

and

$$\begin{aligned} \text{var}(T) &= \int_0^\beta \xi^2 \frac{N\xi^{N-1}}{\beta^N} d\xi - \left(\frac{N\beta}{N+1}\right)^2 \\ &= \frac{N\beta^2}{(N+1)^2(N+2)} \end{aligned}$$

we have that the minimum variance is

$$\text{var}(\hat{\theta}) = \frac{\beta^2}{4N(N+2)}. \tag{5.10}$$

This minimum variance is smaller than that of the sample mean estimator (see (5.9)) for $N \geq 2$ (for $N = 1$ the estimators are identical). The difference in the variances is substantial for large N since the MVU estimator variance goes down as $1/N^2$ as compared to the $1/N$ dependence for the sample mean estimator variance. $\diamond$

The success of the previous example hinges on being able to find a single sufficient statistic. If this is not possible, then this approach cannot be employed. For instance, recalling the phase estimation example (Example 5.4) we required two statistics, $T_1(\mathbf{x})$ and $T_2(\mathbf{x})$. Theorem 5.2 may be extended to address this case. However, the MVU estimator (assuming complete sufficient statistics or $E(g(T_1, T_2)) = 0$ for all ϕ implies that $g = 0$) would need to be determined as $\hat{\phi} = E(\check{\phi}|T_1, T_2)$, where $\check{\phi}$ is any unbiased estimator of the phase. Otherwise, we will need to guess at the form of the function g that combines T_1 and T_2 into a single unbiased estimator for ϕ (see Problem 5.10). As mentioned previously, the evaluation of the conditional expectation is difficult.

5.6 Extension to a Vector Parameter

We now extend our results to search for the MVU estimator for a $p \times 1$ vector parameter $\boldsymbol{\theta}$. In this case we are seeking an unbiased vector estimator (each element is unbiased), such that each element has minimum variance. Many of the results of the previous sections extend naturally. In the vector case we may have more sufficient statistics than the number of parameters ($r > p$), exactly the same number ($r = p$), or even fewer sufficient statistics than parameters to be estimated ($r < p$). The cases of $r > p$ and $r = p$ will be discussed in the following examples. For an example of $r < p$ see Problem 5.16. From our previous discussions a desirable situation would be to have $r = p$. This would allow us to determine the MVU estimator by transforming the sufficient statistics to be unbiased. The RBLS theorem may be extended to justify this approach for a vector parameter. Unfortunately, this approach is not always possible. Before presenting some examples to illustrate these possibilities, extensions of the basic definitions and theorems will be given.

A vector statistic $\mathbf{T}(\mathbf{x}) = [T_1(\mathbf{x})\, T_2(\mathbf{x}) \ldots T_r(\mathbf{x})]^T$ is said to be sufficient for the estimation of $\boldsymbol{\theta}$ if the PDF of the data conditioned on the statistic or $p(\mathbf{x}|\mathbf{T}(\mathbf{x}); \boldsymbol{\theta})$ does not depend on $\boldsymbol{\theta}$. This is to say that $p(\mathbf{x}|\mathbf{T}(\mathbf{x}); \boldsymbol{\theta}) = p(\mathbf{x}|\mathbf{T}(\mathbf{x}))$. In general, many such

$\mathbf{T}(\mathbf{x})$ exist (including the data $\mathbf{x}$), so that we are interested in the *minimal sufficient statistic* or the $\mathbf{T}$ of minimum dimension. To find the minimal sufficient statistic we can again resort to the Neyman-Fisher factorization theorem which for the vector parameter case is as follows.

Theorem 5.3 (Neyman-Fisher Factorization Theorem (Vector Parameter)) *If we can factor the PDF* $p(\mathbf{x};\boldsymbol{\theta})$ *as*

$$p(\mathbf{x};\boldsymbol{\theta}) = g(\mathbf{T}(\mathbf{x}),\boldsymbol{\theta})h(\mathbf{x}) \tag{5.11}$$

where g *is a function depending only on* $\mathbf{x}$ *through* $\mathbf{T}(\mathbf{x})$*, an* $r \times 1$ *statistic, and also on* $\boldsymbol{\theta}$*, and* h *is a function depending only on* $\mathbf{x}$*, then* $\mathbf{T}(\mathbf{x})$ *is a sufficient statistic for* $\boldsymbol{\theta}$*. Conversely, if* $\mathbf{T}(\mathbf{x})$ *is a sufficient statistic for* $\boldsymbol{\theta}$*, then the PDF can be factored as in (5.11).*

A proof can be found in [Kendall and Stuart 1979]. Note that there always exists a set of sufficient statistics, the data set $\mathbf{x}$ itself being sufficient. However, it is the dimensionality of the sufficient statistic that is in question, and that is what the factorization theorem will tell us. Some examples now follow.

Example 5.9 - Sinusoidal Parameter Estimation

Assume that a sinusoidal signal is embedded in WGN

$$x[n] = A\cos 2\pi f_0 n + w[n] \qquad n = 0, 1, \ldots, N-1$$

where the amplitude A, frequency f_0, and noise variance σ^2 are unknown. The unknown parameter vector is therefore $\boldsymbol{\theta} = [A\ f_0\ \sigma^2]^T$. The PDF is

$$p(\mathbf{x};\boldsymbol{\theta}) = \frac{1}{(2\pi\sigma^2)^{\frac{N}{2}}} \exp\left[-\frac{1}{2\sigma^2}\sum_{n=0}^{N-1}(x[n] - A\cos 2\pi f_0 n)^2\right].$$

Expanding the exponent, we have

$$\sum_{n=0}^{N-1}(x[n] - A\cos 2\pi f_0 n)^2 = \sum_{n=0}^{N-1} x^2[n] - 2A\sum_{n=0}^{N-1} x[n]\cos 2\pi f_0 n + A^2\sum_{n=0}^{N-1}\cos^2 2\pi f_0 n.$$

Now because of the term $\sum_{n=0}^{N-1} x[n]\cos 2\pi f_0 n$, where f_0 is *unknown*, we cannot reduce the PDF to have the form expressed in (5.11). If, on the other hand, the frequency were known, the unknown parameter vector would be $\boldsymbol{\theta} = [A\ \sigma^2]^T$ and the PDF could be expressed as

$$p(\mathbf{x};\boldsymbol{\theta}) =$$

$$\underbrace{\frac{1}{(2\pi\sigma^2)^{\frac{N}{2}}}\exp\left[-\frac{1}{2\sigma^2}\left(\sum_{n=0}^{N-1} x^2[n] - 2A\sum_{n=0}^{N-1} x[n]\cos 2\pi f_0 n + A^2\sum_{n=0}^{N-1}\cos^2 2\pi f_0 n\right)\right]}_{g(\mathbf{T}(\mathbf{x}),\boldsymbol{\theta})} \cdot \underbrace{1}_{h(\mathbf{x})}$$

where

$$\mathbf{T}(\mathbf{x}) = \begin{bmatrix} \sum_{n=0}^{N-1} x[n] \cos 2\pi f_0 n \\ \sum_{n=0}^{N-1} x^2[n] \end{bmatrix}.$$

Hence, $\mathbf{T}(\mathbf{x})$ is a sufficient statistic for A and σ^2, but only if f_0 is known (see also Problem 5.17). The next example continues this problem. ◇

Example 5.10 - DC Level in White Noise with Unknown Noise Power

If

$$x[n] = A + w[n] \qquad n = 0, 1, \ldots, N-1$$

where $w[n]$ is WGN with variance σ^2 and the unknown parameters are A and σ^2, we can use the results of the previous example to find a sufficient statistic of dimension 2. Letting the unknown vector parameter be $\boldsymbol{\theta} = [A\,\sigma^2]^T$ and letting $f_0 = 0$ in the previous example, we have as our sufficient statistic

$$\mathbf{T}(\mathbf{x}) = \begin{bmatrix} \sum_{n=0}^{N-1} x[n] \\ \sum_{n=0}^{N-1} x^2[n] \end{bmatrix}.$$

Note that we had already observed that when σ^2 is known, $\sum_{n=0}^{N-1} x[n]$ is a sufficient statistic for A (see Example 5.2), and that when A is known ($A = 0$), $\sum_{n=0}^{N-1} x^2[n]$ is a sufficient statistic for σ^2 (see Example 5.3). In this example, the same statistics are *jointly* sufficient. This will not always be true, however. Since we have two parameters to be estimated and we have found two sufficient statistics, we should be able to find the MVU estimator for this example. ◇

Before actually finding the MVU estimator for the previous example, we need to state the RBLS theorem for vector parameters.

Theorem 5.4 (Rao-Blackwell-Lehmann-Scheffe (Vector Parameter)) *If $\check{\boldsymbol{\theta}}$ is an unbiased estimator of $\boldsymbol{\theta}$ and $\mathbf{T}(\boldsymbol{\theta})$ is an $r \times 1$ sufficient statistic for $\boldsymbol{\theta}$, then $\hat{\boldsymbol{\theta}} = E(\check{\boldsymbol{\theta}}|\mathbf{T}(\mathbf{x}))$ is*

1. *a valid estimator for $\boldsymbol{\theta}$ (not dependent on $\boldsymbol{\theta}$)*
2. *unbiased*
3. *of lesser or equal variance than that of $\check{\boldsymbol{\theta}}$ (each element of $\hat{\boldsymbol{\theta}}$ has lesser or equal variance)*

Additionally, if the sufficient statistic is complete, then $\hat{\boldsymbol{\theta}}$ is the MVU estimator.

In the vector parameter case completeness means that for $\mathbf{v}(\mathbf{T})$, an arbitrary $r \times 1$ function of $\mathbf{T}$, if

$$E(\mathbf{v}(\mathbf{T})) = \int \mathbf{v}(\mathbf{T})p(\mathbf{T};\boldsymbol{\theta})\,d\mathbf{T} = \mathbf{0} \qquad \text{for all } \boldsymbol{\theta} \tag{5.12}$$

then it must be true that

$$\mathbf{v}(\mathbf{T}) = \mathbf{0} \qquad \text{for all } \mathbf{T}.$$

As before, this can be difficult to verify. Also, to be able to determine the MVU estimator directly from $\mathbf{T}(\mathbf{x})$ without having to evaluate $E(\check{\boldsymbol{\theta}}|\mathbf{T}(\mathbf{x}))$, the dimension of the sufficient statistic should be equal to the dimension of the unknown parameter or $r = p$. If this is satisfied, then we need only find an p-dimensional function $\mathbf{g}$ such that

$$E(\mathbf{g}(\mathbf{T})) = \boldsymbol{\theta}$$

assuming that $\mathbf{T}$ is a complete sufficient statistic. We illustrate this approach by continuing Example 5.10.

Example 5.11 - DC Level in WGN with Unknown Noise Power (continued)

We first need to find the mean of the sufficient statistic

$$\mathbf{T}(\mathbf{x}) = \begin{bmatrix} T_1(\mathbf{x}) \\ T_2(\mathbf{x}) \end{bmatrix} = \begin{bmatrix} \sum_{n=0}^{N-1} x[n] \\ \sum_{n=0}^{N-1} x^2[n] \end{bmatrix}.$$

Taking the expected value produces

$$E(\mathbf{T}(\mathbf{x})) = \begin{bmatrix} NA \\ NE(x^2[n]) \end{bmatrix} = \begin{bmatrix} NA \\ N(\sigma^2 + A^2) \end{bmatrix}.$$

Clearly, we could remove the bias of the first component of $\mathbf{T}(\mathbf{x})$ by dividing the statistic by N. However, this would not help the second component. It should be obvious that $T_2(\mathbf{x}) = \sum_{n=0}^{N-1} x^2[n]$ estimates the second moment and not the variance as desired. If we transform $\mathbf{T}(\mathbf{x})$ as

$$\begin{aligned} \mathbf{g}(\mathbf{T}(\mathbf{x})) &= \begin{bmatrix} \frac{1}{N}T_1(\mathbf{x}) \\ \frac{1}{N}T_2(\mathbf{x}) - \left[\frac{1}{N}T_1(\mathbf{x})\right]^2 \end{bmatrix} \\ &= \begin{bmatrix} \bar{x} \\ \frac{1}{N}\sum_{n=0}^{N-1} x^2[n] - \bar{x}^2 \end{bmatrix} \end{aligned}$$

then

$$E(\bar{x}) = A$$

and

$$E\left(\frac{1}{N}\sum_{n=0}^{N-1} x^2[n] - \bar{x}^2\right) = \sigma^2 + A^2 - E(\bar{x}^2).$$

But we know that $\bar{x} \sim \mathcal{N}(A, \sigma^2/N)$, so that

$$E(\bar{x}^2) = A^2 + \sigma^2/N$$

and we have that

$$E\left(\frac{1}{N}\sum_{n=0}^{N-1} x^2[n] - \bar{x}^2\right) = \sigma^2(1 - \frac{1}{N}) = \frac{N-1}{N}\sigma^2.$$

If we multiply this statistic by $N/(N-1)$, it will then be unbiased for σ^2. Finally, the appropriate transformation is

$$\begin{aligned} \mathbf{g}(\mathbf{T}(\mathbf{x})) &= \begin{bmatrix} \frac{1}{N}T_1(\mathbf{x}) \\ \frac{1}{N-1}\left[T_2(\mathbf{x}) - N(\frac{1}{N}T_1(\mathbf{x}))^2\right] \end{bmatrix} \\ &= \begin{bmatrix} \bar{x} \\ \frac{1}{N-1}\left[\sum_{n=0}^{N-1} x^2[n] - N\bar{x}^2\right] \end{bmatrix}. \end{aligned}$$

However, since

$$\begin{aligned} \sum_{n=0}^{N-1}(x[n] - \bar{x})^2 &= \sum_{n=0}^{N-1} x^2[n] - 2\sum_{n=0}^{N-1} x[n]\bar{x} + N\bar{x}^2 \\ &= \sum_{n=0}^{N-1} x^2[n] - N\bar{x}^2, \end{aligned}$$

this may be written as

$$\hat{\boldsymbol{\theta}} = \begin{bmatrix} \bar{x} \\ \frac{1}{N-1}\sum_{n=0}^{N-1}(x[n] - \bar{x})^2 \end{bmatrix}$$

which is the MVU estimator of $\boldsymbol{\theta} = [A\ \sigma^2]^T$. The normalizing factor of $1/(N-1)$ for $\hat{\sigma^2}$ (the sample variance) is due to the one degree of freedom lost in estimating the mean. In this example $\hat{\boldsymbol{\theta}}$ is not efficient. It can be shown [Hoel, Port, and Stone 1971] that $\bar{x}$

and $\hat{\sigma^2}$ are independent and that

$$\begin{aligned} \bar{x} &\sim \mathcal{N}\left(A, \frac{\sigma^2}{N}\right) \\ \frac{(N-1)\hat{\sigma^2}}{\sigma^2} &\sim \chi^2_{N-1}. \end{aligned}$$

Hence, the covariance matrix is

$$\mathbf{C}_{\hat{\theta}} = \begin{bmatrix} \frac{\sigma^2}{N} & 0 \\ 0 & \frac{2\sigma^4}{N-1} \end{bmatrix}$$

while the CRLB is (see Example 3.6)

$$\mathbf{I}^{-1}(\boldsymbol{\theta}) = \begin{bmatrix} \frac{\sigma^2}{N} & 0 \\ 0 & \frac{2\sigma^4}{N} \end{bmatrix}.$$

Thus, the MVU estimator could not have been obtained by examining the CRLB.

Finally, we should observe that if we had known beforehand or suspected that the sample mean and sample variance were the MVU estimators, then we could have reduced our work. The PDF

$$p(\mathbf{x};\boldsymbol{\theta}) = \frac{1}{(2\pi\sigma^2)^{\frac{N}{2}}} \exp\left[-\frac{1}{2\sigma^2}\sum_{n=0}^{N-1}(x[n]-A)^2\right]$$

can be factored more directly by noting that

$$\begin{aligned} \sum_{n=0}^{N-1}(x[n]-A)^2 &= \sum_{n=0}^{N-1}(x[n]-\bar{x}+\bar{x}-A)^2 \\ &= \sum_{n=0}^{N-1}(x[n]-\bar{x})^2 + 2(\bar{x}-A)\sum_{n=0}^{N-1}(x[n]-\bar{x}) + N(\bar{x}-A)^2. \end{aligned}$$

The middle term is zero, so that we have the factorization

$$p(\mathbf{x};\boldsymbol{\theta}) = \underbrace{\frac{1}{(2\pi\sigma^2)^{\frac{N}{2}}} \exp\left\{-\frac{1}{2\sigma^2}\left[\sum_{n=0}^{N-1}(x[n]-\bar{x})^2 + N(\bar{x}-A)^2\right]\right\}}_{g(\mathbf{T}'(\mathbf{x}),\boldsymbol{\theta})} \cdot \underbrace{1}_{h(\mathbf{x})}$$

where

$$\mathbf{T}'(\mathbf{x}) = \begin{bmatrix} \bar{x} \\ \sum_{n=0}^{N-1}(x[n]-\bar{x})^2 \end{bmatrix}.$$

Dividing the second component by $(N-1)$ would produce $\hat{\boldsymbol{\theta}}$. Of course $\mathbf{T}'(\mathbf{x})$ and $\mathbf{T}(\mathbf{x})$ are related by a one-to-one transformation, illustrating once again that the sufficient statistic is unique to within these transformations.

In asserting that $\hat{\boldsymbol{\theta}}$ is the MVU estimator we have not verified the completeness of the statistic. Completeness follows because the Gaussian PDF is a special case of the vector exponential family of PDFs, which are known to be complete [Kendall and Stuart 1979] (see Problem 5.14 for the definition of the scalar exponential family of PDFs). ◇

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Problems

5.1 For Example 5.3 prove that $\sum_{n=0}^{N-1} x^2[n]$ is a sufficient statistic for σ^2 by using the definition that $p(\mathbf{x}|\sum_{n=0}^{N-1} x^2[n] = T_0; \sigma^2)$ does not depend on σ^2. Hint: Note that the PDF of $s = \sum_{n=0}^{N-1} x^2[n]/\sigma^2$ is a chi-squared distribution with N degrees of freedom or

$$p(s) = \begin{cases} \frac{1}{2^{\frac{N}{2}}\Gamma(\frac{N}{2})} \exp\left(-\frac{s}{2}\right) s^{\frac{N}{2}-1} & s > 0 \\ 0 & s < 0. \end{cases}$$

5.2 The IID observations $x[n]$ for $n = 0, 1, \ldots, N-1$ have the Rayleigh PDF

$$p(x[n]; \sigma^2) = \begin{cases} \frac{x[n]}{\sigma^2} \exp\left(-\frac{1}{2}\frac{x^2[n]}{\sigma^2}\right) & x[n] > 0 \\ 0 & x[n] < 0. \end{cases}$$

Find a sufficient statistic for σ^2.

5.3 The IID observations $x[n]$ for $n = 0, 1, \ldots, N-1$ have the exponential PDF

$$p(x[n]; \lambda) = \begin{cases} \lambda \exp(-\lambda x[n]) & x[n] > 0 \\ 0 & x[n] < 0. \end{cases}$$

Find a sufficient statistic for λ.

5.4 The IID observations $x[n]$ for $n = 0, 1, \ldots, N-1$ are distributed according to $\mathcal{N}(\theta, \theta)$, where $\theta > 0$. Find a sufficient statistic for θ.

5.5 The IID observations $x[n]$ for $n = 0, 1, \ldots, N-1$ are distributed according to $\mathcal{U}[-\theta, \theta]$, where $\theta > 0$. Find a sufficient statistic for θ.

5.6 If $x[n] = A + w[n]$ for $n = 0, 1, \ldots, N-1$ are observed, where $w[n]$ is WGN with variance σ^2, find the MVU estimator for σ^2 assuming that A is known. You may assume that the sufficient statistic is complete.

5.7 Consider the frequency estimation of a sinusoid embedded in WGN or

$$x[n] = \cos 2\pi f_0 n + w[n] \qquad n = 0, 1, \ldots, N-1$$

where $w[n]$ is WGN of known variance σ^2. Show that it is not possible to find a sufficient statistic for the frequency f_0.

5.8 In a similar fashion to Problem 5.7 consider the estimation of the damping constant r for the data

$$x[n] = r^n + w[n] \qquad n = 0, 1, \ldots, N-1$$

where $w[n]$ is WGN with known variance σ^2. Again show that a sufficient statistic does not exist for r.

5.9 Assume that $x[n]$ is the result of a Bernoulli trial (a coin toss) with

$$\begin{aligned} \Pr\{x[n] = 1\} &= \theta \\ \Pr\{x[n] = 0\} &= 1 - \theta \end{aligned}$$

and that N IID observations have been made. Assuming the Neyman-Fisher factorization theorem holds for discrete random variables, find a sufficient statistic for θ. Then, assuming completeness, find the MVU estimator of θ.

5.10 For Example 5.4 the following estimator is proposed

$$\hat{\phi} = -\arctan\left[\frac{T_2(\mathbf{x})}{T_1(\mathbf{x})}\right].$$

Show that at high SNR or $\sigma^2 \to 0$, for which

$$\begin{aligned} T_1(\mathbf{x}) &\approx \sum_{n=0}^{N-1} A\cos(2\pi f_0 n + \phi)\cos 2\pi f_0 n \\ T_2(\mathbf{x}) &\approx \sum_{n=0}^{N-1} A\cos(2\pi f_0 n + \phi)\sin 2\pi f_0 n \end{aligned}$$

the estimator satisfies $\hat{\phi} \approx \phi$. Use any approximations you deem appropriate. Is the proposed estimator the MVU estimator?

5.11 For Example 5.2 it is desired to estimate $\theta = 2A + 1$ instead of A. Find the MVU estimator of θ. What if the parameter $\theta = A^3$ is desired? Hint: Reparameterize the PDF in terms of θ and use the standard approach.

5.12 This problem examines the idea that sufficient statistics are unique to within one-to-one transformations. Once again, consider Example 5.2 but with the sufficient statistics

$$\begin{aligned} T_1(\mathbf{x}) &= \sum_{n=0}^{N-1} x[n] \\ T_2(\mathbf{x}) &= \left(\sum_{n=0}^{N-1} x[n] \right)^3. \end{aligned}$$

Find functions g_1 and g_2 so that the unbiased condition is satisfied or

$$\begin{aligned} E[g_1(T_1(\mathbf{x}))] &= A \\ E[g_2(T_2(\mathbf{x}))] &= A \end{aligned}$$

and hence the MVU estimator. Is the transformation from T_1 to T_2 one-to-one? What would happen if the statistic $T_3(\mathbf{x}) = (\sum_{n=0}^{N-1} x[n])^2$ were considered? Could you find a function g_3 to satisfy the unbiased constraint?

5.13 In this problem we derive the MVU estimator for an example that we have not encountered previously. If N IID observations are made according to the PDF

$$p(x[n];\theta) = \begin{cases} \exp[-(x[n]-\theta)] & x[n] > \theta \\ 0 & x[n] < \theta, \end{cases}$$

find the MVU estimator for θ. Note that θ represents the minimum value that $x[n]$ may attain. Assume that the sufficient statistic is complete.

5.14 For the random variable x consider the PDF

$$p(x;\theta) = \exp\left[A(\theta)B(x) + C(x) + D(\theta)\right].$$

A PDF of this type is said to belong to the *scalar exponential family* of PDFs. Many useful properties are associated with this family, and in particular, that of sufficient statistics, which is explored in this and the next problem. Show that the following PDFs are within this class.

a. Gaussian

$$p(x;\mu) = \frac{1}{\sqrt{2\pi}} \exp\left[-\frac{1}{2}(x-\mu)^2\right].$$

b. Rayleigh

$$p(x;\sigma^2) = \begin{cases} \dfrac{x}{\sigma^2} \exp\left[-\dfrac{1}{2}\dfrac{x^2}{\sigma^2}\right] & x > 0 \\ 0 & x < 0. \end{cases}$$

c. Exponential

$$p(x;\lambda) = \begin{cases} \lambda \exp[-\lambda x] & x > 0 \\ 0 & x < 0 \end{cases}$$

where $\lambda > 0$.

5.15 If we observe $x[n]$ for $n = 0, 1, \ldots, N-1$, which are IID and whose PDF belongs to the exponential family of PDFs, show that

$$T(\mathbf{x}) = \sum_{n=0}^{N-1} B(x[n])$$

is a sufficient statistic for θ. Next apply this result to determine the sufficient statistic for the Gaussian, Rayleigh, and exponential PDFs in Problem 5.14. Finally, assuming that the sufficient statistics are complete, find the MVU estimator in each case (if possible). (You may want to compare your results to Example 5.2 and Problems 5.2 and 5.3, respectively.)

5.16 In this problem an example is given where there are fewer sufficient statistics than parameters to be estimated [Kendall and Stuart 1979]. Assume that $x[n]$ for $n = 0, 1, \ldots, N-1$ are observed, where $\mathbf{x} \sim \mathcal{N}(\boldsymbol{\mu}, \mathbf{C})$ and

$$\begin{aligned} \boldsymbol{\mu} &= \begin{bmatrix} N\mu \\ 0 \\ 0 \\ \vdots \\ 0 \end{bmatrix} \\ \mathbf{C} &= \begin{bmatrix} N-1+\sigma^2 & -\mathbf{1}^T \\ -\mathbf{1} & \mathbf{I} \end{bmatrix}. \end{aligned}$$

The covariance matrix $\mathbf{C}$ has the dimensions

$$\begin{bmatrix} 1 \times 1 & 1 \times (N-1) \\ (N-1) \times 1 & (N-1) \times (N-1) \end{bmatrix}.$$

Show that with $\boldsymbol{\theta} = [\mu \; \sigma^2]^T$

$$p(\mathbf{x}; \boldsymbol{\theta}) = \frac{1}{(2\pi)^{\frac{N}{2}}\sigma} \exp\left\{-\frac{1}{2}\left[\frac{N^2}{\sigma^2}(\bar{x}-\mu)^2 + \sum_{n=1}^{N-1} x^2[n]\right]\right\}.$$

What are the sufficient statistics for $\boldsymbol{\theta}$? Hint: You will need to find the inverse and determinant of a partitioned matrix and to use Woodbury's identity.

5.17 Consider a sinusoid of known frequency embedded in WGN or

$$x[n] = A\cos 2\pi f_0 n + w[n] \qquad n = 0, 1, \ldots, N-1$$

where $w[n]$ is WGN with variance σ^2. Find the MVU estimator of

a. the amplitude A, assuming σ^2 is known

b. the amplitude A and noise variance σ^2.

You may assume that the sufficient statistics are complete. Hint: The results of Examples 5.9 and 5.11 may prove helpful.

5.18 If $x[n]$ for $n = 0, 1, \ldots, N-1$ are observed, where the samples are IID and distributed according to $\mathcal{U}[\theta_1, \theta_2]$, find a sufficient statistic for $\boldsymbol{\theta} = [\theta_1\ \theta_2]^T$.

5.19 Recall the linear model in Chapter 4 in which the data are modeled as

$$\mathbf{x} = \mathbf{H}\boldsymbol{\theta} + \mathbf{w}$$

where $\mathbf{w} \sim \mathcal{N}(\mathbf{0}, \sigma^2\mathbf{I})$ with σ^2 known. Find the MVU estimator of $\boldsymbol{\theta}$ by using the Neyman-Fisher and RBLS theorems. Assume the sufficient statistic is complete. Compare your result with that described in Chapter 4. Hint: First prove that the following identity holds:

$$(\mathbf{x} - \mathbf{H}\boldsymbol{\theta})^T(\mathbf{x} - \mathbf{H}\boldsymbol{\theta}) = (\mathbf{x} - \mathbf{H}\hat{\boldsymbol{\theta}})^T(\mathbf{x} - \mathbf{H}\hat{\boldsymbol{\theta}}) + (\boldsymbol{\theta} - \hat{\boldsymbol{\theta}})^T\mathbf{H}^T\mathbf{H}(\boldsymbol{\theta} - \hat{\boldsymbol{\theta}})$$

where

$$\hat{\boldsymbol{\theta}} = (\mathbf{H}^T\mathbf{H})^{-1}\mathbf{H}^T\mathbf{x}.$$

Appendix 5A

Proof of Neyman-Fisher Factorization Theorem (Scalar Parameter)

Consider the joint PDF $p(\mathbf{x}, T(\mathbf{x}); \theta)$. In evaluating this PDF it should be noted that $T(\mathbf{x})$ is functionally dependent on $\mathbf{x}$. Hence, the joint PDF must be zero when evaluated at $\mathbf{x} = \mathbf{x}_0$, $T(\mathbf{x}) = T_0$, unless $T(\mathbf{x}_0) = T_0$. This is analogous to the situation where we have two random variables x and y with $x = y$ always. The joint PDF $p(x, y)$ is $p(x)\delta(y - x)$, a line impulse in the two-dimensional plane, and is a degenerate two-dimensional PDF. In our case we have $p(\mathbf{x}, T(\mathbf{x}) = T_0; \theta) = p(\mathbf{x}; \theta)\delta(T(\mathbf{x}) - T_0)$, which we will use in our proof. A second result that we will need is a shorthand way of representing the PDF of a function of several random variables. If $y = g(\mathbf{x})$, for $\mathbf{x}$ a vector random variable, the PDF of y can be written as

$$p(y) = \int p(\mathbf{x})\delta(y - g(\mathbf{x}))\, d\mathbf{x}. \tag{5A.1}$$

Using properties of the Dirac delta function [Hoskins 1979], this can be shown to be equivalent to the usual formula for transformation of random variables [Papoulis 1965]. If, for example, $\mathbf{x}$ is a scalar random variable, then

$$\delta(y - g(x)) = \sum_{i=1}^{k} \frac{\delta(x - x_i)}{\left|\frac{dg}{dx}\right|_{x=x_i}}$$

where $\{x_1, x_2, \ldots, x_k\}$ are all the solutions of $y = g(x)$. Substituting this into (5A.1) and evaluating produces the usual formula.

We begin by proving that $T(\mathbf{x})$ is a sufficient statistic when the factorization holds.

$$\begin{aligned} p(\mathbf{x}|T(\mathbf{x}) = T_0; \theta) &= \frac{p(\mathbf{x}, T(\mathbf{x}) = T_0; \theta)}{p(T(\mathbf{x}) = T_0; \theta)} \\ &= \frac{p(\mathbf{x}; \theta)\delta(T(\mathbf{x}) - T_0)}{p(T(\mathbf{x}) = T_0; \theta)}. \end{aligned}$$

Using the factorization

$$p(\mathbf{x}|T(\mathbf{x}) = T_0; \theta) = \frac{g(T(\mathbf{x}) = T_0, \theta)h(\mathbf{x})\delta(T(\mathbf{x}) - T_0)}{p(T(\mathbf{x}) = T_0; \theta)}. \tag{5A.2}$$

But, from (5A.1),

$$p(T(\mathbf{x}) = T_0; \theta) = \int p(\mathbf{x}; \theta)\delta(T(\mathbf{x}) - T_0)\, d\mathbf{x}.$$

Again using the factorization

$$\begin{aligned} p(T(\mathbf{x}) = T_0; \theta) &= \int g(T(\mathbf{x}) = T_0, \theta)h(\mathbf{x})\delta(T(\mathbf{x}) - T_0)\, d\mathbf{x} \\ &= g(T(\mathbf{x}) = T_0, \theta) \int h(\mathbf{x})\delta(T(\mathbf{x}) - T_0)\, d\mathbf{x}. \end{aligned}$$

The latter step is possible because the integral is zero except over the surface in R^N for which $T(\mathbf{x}) = T_0$. Over this surface g is a constant. Using this in (5A.2) produces

$$p(\mathbf{x}|T(\mathbf{x}) = T_0; \theta) = \frac{h(\mathbf{x})\delta(T(\mathbf{x}) - T_0)}{\int h(\mathbf{x})\delta(T(\mathbf{x}) - T_0)\, d\mathbf{x}}$$

which does not depend on θ. Hence, we conclude that $T(\mathbf{x})$ is a sufficient statistic.

Next we prove that if $T(\mathbf{x})$ is a sufficient statistic, then the factorization holds. Consider the joint PDF

$$p(\mathbf{x}, T(\mathbf{x}) = T_0; \theta) = p(\mathbf{x}|T(\mathbf{x}) = T_0; \theta)p(T(\mathbf{x}) = T_0; \theta) \tag{5A.3}$$

and note that

$$p(\mathbf{x}, T(\mathbf{x}) = T_0; \theta) = p(\mathbf{x}; \theta)\delta(T(\mathbf{x}) - T_0).$$

Because $T(\mathbf{x})$ is a sufficient statistic, the conditional PDF does not depend on θ. Hence, we write it as $p(\mathbf{x}|T(\mathbf{x}) = T_0)$. Now since we are given $T(\mathbf{x}) = T_0$, which defines a surface in R^N, the conditional PDF is nonzero only on that surface. We can let $p(\mathbf{x}|T(\mathbf{x}) = T_0) = w(\mathbf{x})\delta(T(\mathbf{x}) - T_0)$, where

$$\int w(\mathbf{x})\delta(T(\mathbf{x}) - T_0)\, d\mathbf{x} = 1. \tag{5A.4}$$

Thus, substituting in (5A.3) produces

$$p(\mathbf{x}; \theta)\delta(T(\mathbf{x}) - T_0) = w(\mathbf{x})\delta(T(\mathbf{x}) - T_0)p(T(\mathbf{x}) = T_0; \theta). \tag{5A.5}$$

We can let

$$w(\mathbf{x}) = \frac{h(\mathbf{x})}{\int h(\mathbf{x})\delta(T(\mathbf{x}) - T_0)\, d\mathbf{x}}$$

so that (5A.4) is satisfied. Then, (5A.5) becomes

$$p(\mathbf{x};\theta)\delta(T(\mathbf{x})-T_0)=\frac{h(\mathbf{x})\delta(T(\mathbf{x})-T_0)}{\int h(\mathbf{x})\delta(T(\mathbf{x})-T_0)\,d\mathbf{x}}p(T(\mathbf{x})=T_0;\theta)$$

or

$$p(\mathbf{x};\theta)=g(T(\mathbf{x})=T_0;\theta)h(\mathbf{x})$$

where

$$g(T(\mathbf{x})=T_0;\theta)=\frac{p(T(\mathbf{x})=T_0;\theta)}{\int h(\mathbf{x})\delta(T(\mathbf{x})-T_0)\,d\mathbf{x}}.$$

In this form we see how, in principle, the PDF of the sufficient statistic can be found based on the factorization since

$$p(T(\mathbf{x})=T_0;\theta)=g(T(\mathbf{x})=T_0;\theta)\int h(\mathbf{x})\delta(T(\mathbf{x})-T_0)\,d\mathbf{x}.$$

Appendix 5B

Proof of Rao-Blackwell-Lehmann-Scheffe Theorem (Scalar Parameter)

To prove (1) of Theorem 5.2, that $\hat{\theta}$ is a valid estimator of θ (not a function of θ), note that

$$\begin{aligned} \hat{\theta} &= E(\check{\theta}|T(\mathbf{x})) \\ &= \int \check{\theta}(\mathbf{x}) p(\mathbf{x}|T(\mathbf{x});\theta)\, d\mathbf{x}. \end{aligned} \tag{5B.1}$$

By the definition of a sufficient statistic, $p(\mathbf{x}|T(\mathbf{x});\theta)$ does not depend on θ but only on $\mathbf{x}$ and $T(\mathbf{x})$. Therefore, after the integration is performed over $\mathbf{x}$, the result will be solely a function of T and therefore of $\mathbf{x}$ only.

To prove (2), that $\hat{\theta}$ is unbiased, we use (5B.1) and the fact that $\hat{\theta}$ is a function of T only.

$$\begin{aligned} E(\hat{\theta}) &= \int\int \check{\theta}(\mathbf{x}) p(\mathbf{x}|T(\mathbf{x});\theta)\, d\mathbf{x}\, p(T(\mathbf{x});\theta)\, dT \\ &= \int \check{\theta}(\mathbf{x}) \int p(\mathbf{x}|T(\mathbf{x});\theta) p(T(\mathbf{x});\theta)\, dT\, d\mathbf{x} \\ &= \int \check{\theta}(\mathbf{x}) p(\mathbf{x};\theta)\, d\mathbf{x} \\ &= E(\check{\theta}). \end{aligned}$$

But $\check{\theta}$ is unbiased by assumption, and therefore

$$E(\hat{\theta}) = E(\check{\theta}) = \theta.$$

To prove (3), that

$$\mathrm{var}(\hat{\theta}) \leq \mathrm{var}(\check{\theta})$$

we have

$$\begin{aligned}\text{var}(\check{\theta}) &= E\left[(\check{\theta} - E(\check{\theta}))^2\right] \\ &= E\left[(\check{\theta} - \hat{\theta} + \hat{\theta} - \theta)^2\right] \\ &= E[(\check{\theta} - \hat{\theta})^2] + 2E[(\check{\theta} - \hat{\theta})(\hat{\theta} - \theta)] + E[(\hat{\theta} - \theta)^2].\end{aligned}$$

The cross-term in the above expression, $E[(\check{\theta} - \hat{\theta})(\hat{\theta} - \theta)]$, is now shown to be zero. We know that $\hat{\theta}$ is solely a function of T. Thus, the expectation of the cross-term is with respect to the joint PDF of T and $\check{\theta}$ or

$$E_{T,\check{\theta}}[(\check{\theta} - \hat{\theta})(\hat{\theta} - \theta)] = E_T E_{\check{\theta}|T}[(\check{\theta} - \hat{\theta})(\hat{\theta} - \theta)].$$

But

$$E_{\check{\theta}|T}[(\check{\theta} - \hat{\theta})(\hat{\theta} - \theta)] = E_{\check{\theta}|T}[\check{\theta} - \hat{\theta}](\hat{\theta} - \theta)$$

and also

$$E_{\check{\theta}|T}[\check{\theta} - \hat{\theta}] = E_{\check{\theta}|T}(\check{\theta}|T) - \hat{\theta} = \hat{\theta} - \hat{\theta} = 0.$$

Thus, we have the desired result

$$\begin{aligned}\text{var}(\check{\theta}) &= E[(\check{\theta} - \hat{\theta})^2] + \text{var}(\hat{\theta}) \\ &\geq \text{var}(\hat{\theta}).\end{aligned}$$

Finally, that $\hat{\theta}$ is the MVU estimator if it is complete follows from the discussion given in Section 5.5.